

Supporting Information: Auditing Whether Models Learn RNA Sequence or Database Structure in RNALocate

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Supplementary Methods

Frozen analysis plan

The context-shortcut audit was frozen on 14 August 2026 before the new analyses were run. Existing upstream annotation-recovery results were already known, so the plan is a supplemental post hoc analysis plan rather than a preregistration.

Bootstrap

Average-precision intervals and contrasts used 1,000 paired resamples of all-sequence graph components. Component resampling preserves within-component dependence and repeated contexts belonging to a gene.

Support-gradient intervals

Observed fractions at support values 0-15 used Wilson 95% binomial intervals. Support zero is a cohort-selection boundary and is not interpreted as the first point of a causal dose-response curve.

Permutation control

Each of 200 replicates permuted support among genes within each context and split, repeated hyperparameter selection, refit on training plus validation, and evaluated held-out components.

Calibration

Context, context-plus-support, and full-model predictions were separated into ten equal-frequency test bins. Predicted probability was compared with observed fraction and Wilson intervals.

Sequence-view sensitivity

Composition plus log length, normalized 3-mer frequencies, and the complete 70-feature representation were evaluated under the same partition, tuning grid, and logistic-regression family.

Supplementary Tables

Table S1 | Context-specific context-plus-support performance

Context	Pairs	Prev.	AP	AP lift	AUROC	Brier
A549	2822	0.7101	0.9420	0.2318	0.9041	0.1011
GM12878	2822	0.7381	0.9462	0.2081	0.8871	0.1160
H1-hESC	2822	0.8795	0.9901	0.1105	0.9409	0.0612
HEK293T	2822	0.9745	0.9922	0.0177	0.8067	0.0258
HT1080	2822	0.6687	0.9662	0.2975	0.9464	0.0877
HUVEC	2822	0.7158	0.9783	0.2625	0.9561	0.0743
HeLa-S3	2822	0.6421	0.9205	0.2784	0.8983	0.1180
HepG2	2822	0.7155	0.9655	0.2501	0.9275	0.0986
IMR-90	2822	0.3629	0.4768	0.1139	0.6740	0.2290
K562	2822	0.6545	0.9226	0.2681	0.8899	0.1258
MCF-7	2822	0.7555	0.9752	0.2198	0.9350	0.0881
NCI-H460	2822	0.5595	0.8751	0.3155	0.8977	0.1165
NHEK	2822	0.7300	0.9730	0.2430	0.9381	0.0885
SK-MEL-5	2822	0.5829	0.8974	0.3145	0.9028	0.1168
SK-N-DZ	2822	0.6244	0.8959	0.2716	0.8819	0.1265
SK-N-SH	2822	0.6882	0.9201	0.2320	0.8756	0.1232

Table S2 | Low-level sequence representation sensitivity

Feature view	Features	Model	C	AP	AUROC	Brier
composition_length	6	sequence_only	10.0000	0.7870	0.6689	0.1963
composition_length	6	full	0.0100	0.9577	0.9156	0.1059
three_mer_only	64	sequence_only	1.0000	0.8139	0.7034	0.1890
three_mer_only	64	full	0.0100	0.9580	0.9158	0.1059
complete_70	70	sequence_only	10.0000	0.8234	0.7153	0.1854
complete_70	70	full	0.0100	0.9580	0.9158	0.1058

Table S3 | Incremental AP effects

Contrast	AP difference	CI low	CI high	Replicates
sequence_after_context	0.0787	0.0719	0.0863	1000
support_after_context	0.1535	0.1458	0.1616	1000
sequence_after_context_support	0.0009	0.0007	0.0010	1000
support_after_context_sequence	0.0757	0.0695	0.0823	1000

Table S4 | Annotation-recovery split sensitivity

Partition	Method	Comp.	Largest	Context AP	3-mer AP	C+S AP	Full AP
primary_g80_c80	exhaustive threshold graph	18437	14	0.7198	0.3054	0.9572	0.9580
cluster_g70_c80	MMseqs2 easy-cluster stress test	18120	21	0.7205	0.3055	0.9576	0.9584
cluster_g60_c80	MMseqs2 easy-cluster stress test	17887	21	0.7201	0.3052	0.9570	0.9579
cluster_g80_c50	MMseqs2 easy-cluster stress test	18019	15	0.7197	0.3054	0.9576	0.9585

Table S5 | Support-zero exclusion sensitivity

Model	Pairs	Prev.	AP	CI low	CI high	AUROC	Brier
Context	45003	0.6866	0.8014	0.7923	0.8115	0.6560	0.1987
Other-context support	45003	0.6866	0.9006	0.8959	0.9051	0.8558	0.1302
Context + support	45003	0.6866	0.9573	0.9544	0.9601	0.9159	0.1059
Context + support + sequence	45003	0.6866	0.9582	0.9552	0.9610	0.9164	0.1057